Final Review Reading Guide

Design of a Temperature-Compensated PLL Frequency Synthesiser

For Viju Sanjai — Final Review (Review III), 28.10.2026
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This is a reading document. It explains what was done, how to read every result plot, what each number means, and how to answer an examiner. All numbers come from the generated results files in the repository (`results/final_numbers.md`, `results/corner_audit/FINDINGS.md`, `results/comp/mc_v2_analysis.txt`, `results/loop/campaign/summary.csv`). Several Review 2 claims were corrected after Review 2; those old claims appear *only* in Section 10. If an examiner quotes an old number from the Review 2 slides, check that section first.

Contents

1 The story in 60 seconds **3**

2 The system and each block **3**
2.1 K_{VCO} and the low-voltage region . . . 4

3 Uncompensated drift: the problem we are solving **5**
3.1 Why the drift is curved . . . 5

4 Corner audit: why sf/fs numbers looked out of range **6**
4.1 Process corners from scratch . . . 6
4.2 Why NMOS drive sets this VCO . . . 6
4.3 The PDK skew corners are wider than ss/ff . . . 6
4.4 The textbook 3×3 corner set . . . 7
4.5 Physics sanity checks (all pass) . . . 8

5 VCO characterisation and the tempco mechanism **8**

6 The compensation network and how the fit works **9**
6.1 Typical corner . . . 9
6.2 All 11 corners . . . 10

7 Monte Carlo, $n = 72$ **11**
7.1 What a paired test is, in plain language . . . 11
7.2 Results . . . 12
7.3 The mismatch floor, and why this is an honest finding . . . 12

8 Closed loop **12**
8.1 Loop dynamics, found, fixed, verified . . . 12
8.2 Cold start . . . 13
8.3 Lock across 11 corners . . . 13
8.4 Temperature, and the ss/85 °C miss . . . 14
8.5 Closed-loop vs open-loop voltage check . . . 15

9 Method, novelty and limitations **15**

10 Corrections log **17**

11 Numbers to memorise **18**

12 Likely examiner questions and model answers **18**

13 Glossary **20**

1 The story in 60 seconds

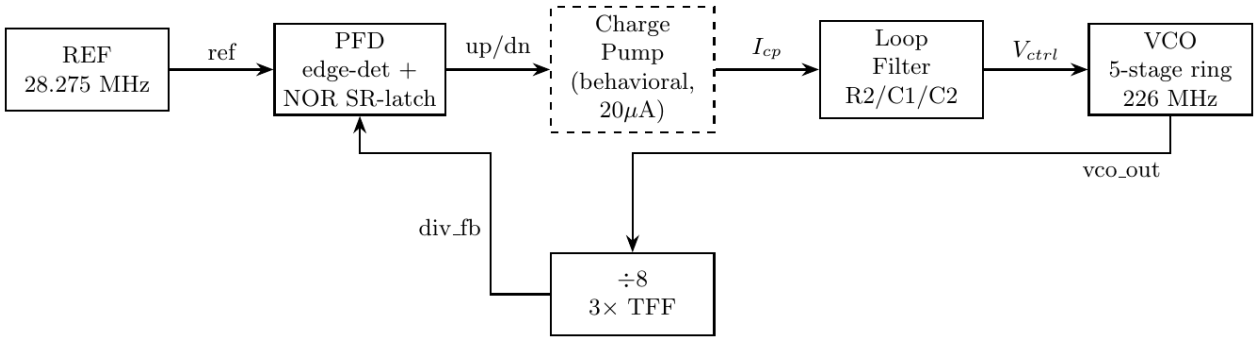
Say this

A ring-oscillator VCO changes frequency with temperature, so a PLL built around it has to work harder as the chip heats up. We built a complete integer-N charge-pump PLL in the open sky130B process and simulated it in ngspice. The VCO drifts 8.5% over −40 to 85 °C at the typical corner, and between 1.5% and 17.5% across 11 process corners. We add a correction voltage $k_1(T-27)+k_2(T-27)^2$ to the VCO control voltage. Fitted properly on the measured frequency surface and verified at held-out temperatures, the 2nd-order version cuts the typical-corner span from 8.51% to 0.20% and meets our ±0.3% target at 8 of 11 corners; the other three are 0.30, 0.33 and 0.36%, just outside. With transistor mismatch (72 virtual chips) the correction removes about 17 MHz of the 20 MHz drift, leaving about 3 MHz (1.2–1.4%): a mismatch floor that one global correction cannot remove. Here 2nd order is not statistically better than 1st order, which is an honest and correct finding. The closed loop (retuned to $\zeta \approx 0.70$) locks to $8 \times f_{ref} = 226.17$ MHz at all 11 corners from both a coarse and a true 0 V start, inside the 2 µs spec, with one honest exception: ss at 85 °C takes 2083 ns, 4% over.

What the final review adds over Review 2

1. **Corner audit:** we were asked why sf/fs numbers look out of range; we investigated and found the PDK's skew corners are simply wider than ss/ff. We added a textbook 3 × 3 corner set and report 11 corners.
2. **Corrected compensation:** the earlier coefficients were limited by a wrong K_{VCO}; the new fit meets spec at tt and 8/11 corners.
3. **Corrected Monte Carlo:** $n = 72$ with the corrected coefficients.
4. **Full closed-loop campaign:** 28 transistor-level runs across 11 corners and temperatures.

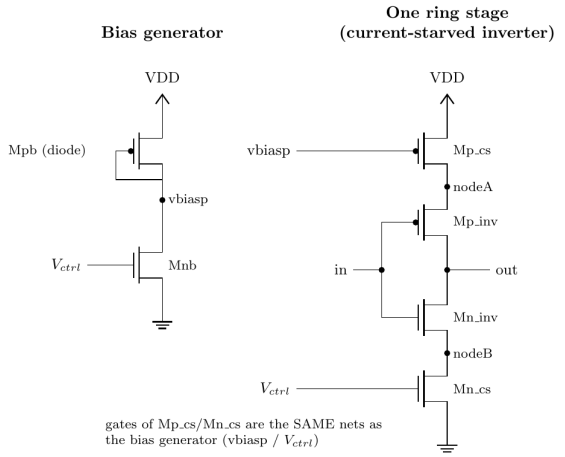
2 The system and each block

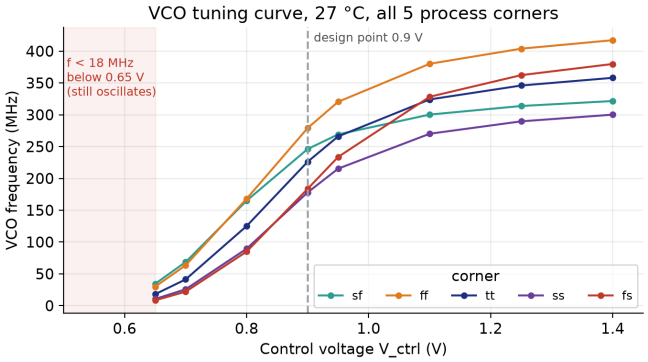


Closed-loop lock (tt, 27°C): **503 ns** (R2 = 3.6 kΩ; spec <2 µs)

- **Reference:** 28.275 MHz. Output is 8×, 226.2 MHz, at $V_{ctrl} = 0.9$ V, tt, 27 °C.
- **PFD (phase-frequency detector):** compares reference and divided-VCO edges and produces UP/DN pulses proportional to the phase error. Built from edge detectors and NOR SR-latches (resettable D flip-flops failed because the asynchronous reset fought the latch feedback in our hand-built cells).
- **Charge pump:** UP/DN pulses become ±20 µA current pulses into the loop filter. It is a *behavioural* (ideal) model, not transistors. It runs at 5× until the first divider edge (cold-start boost).

- **Loop filter:** $R_2 = 3.6$ kΩ in series with $C_1 = 68$ pF, with $C_2 = 6.8$ pF in parallel. It integrates current into V_{ctrl}. R_2 adds the zero that stabilises the loop; C_2 smooths ripple. (R_2 was 1.7 kΩ in Reviews 1–2.)
- **VCO:** a 5-stage current-starved ring. V_{ctrl} sets the starving current, which sets the delay per stage and therefore the frequency.
- **Divider ÷8:** three toggle flip-flops, feeding the divided clock back to the PFD.





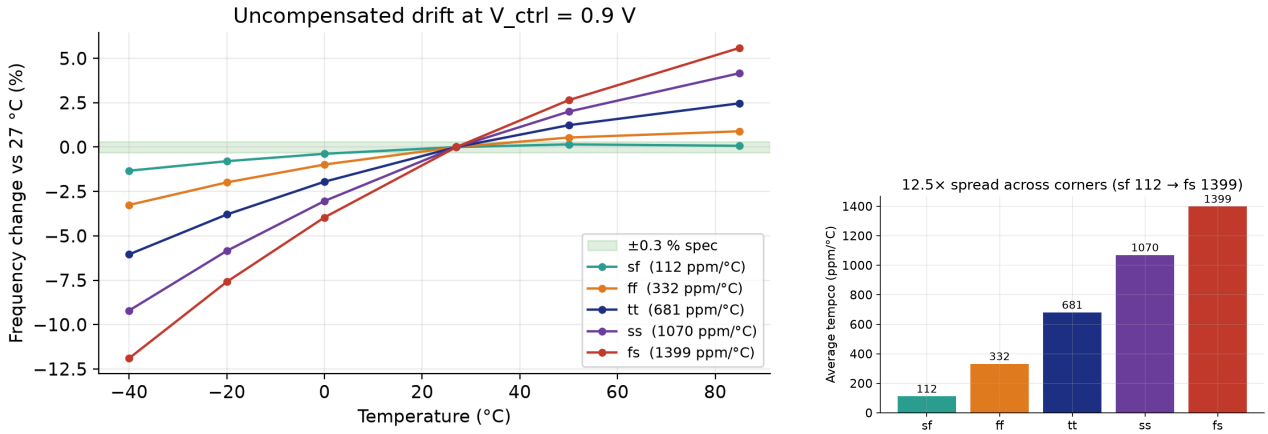
How to read this figure

Axes: x: control voltage V_{ctrl} (V); y: VCO frequency (MHz); one curve per PDK corner at 27 °C; dashed line is the 0.9 V design point.
Good vs bad: A good tuning curve is monotonic and covers the target (226 MHz) well inside its range, which it does. The slope is not constant (S-shape): steep around 0.8–1.0 V, flat above 1.2 V. The curves fan out with process: at 0.9 V ss 178.0 < fs 183.7 < tt 226.2 < sf 246.2 < ff 279.2 MHz.
Key number: tt tunes 18 MHz at 0.65 V to 358 MHz at 1.40 V; local gain $K_{VCO} \approx 890$ MHz/V at 0.9 V (tt).
Takeaway: One control voltage covers 226 MHz at every corner, but the voltage needed differs by corner (0.85–0.97 V), which later explains the lock voltages.

2.1 K_{VCO} and the low-voltage region

$K_{VCO} = df/dV_{ctrl}$ is measured as a central difference at 0.9 ± 0.05 V: tt 887 (about 890) MHz/V; sf/ff/ss/fs: 580/940/820/1020 MHz/V. There is no hard oscillation floor: the ring keeps running in subthreshold (0.63 MHz at 0.5 V, 6.5 MHz at 0.6 V, 18 MHz at 0.65 V). But below about 0.65 V it is under 8% of the target, so from a 0 V start the divided clock is effectively absent; this is why the cold start needs the boost.

3 Uncompensated drift: the problem we are solving

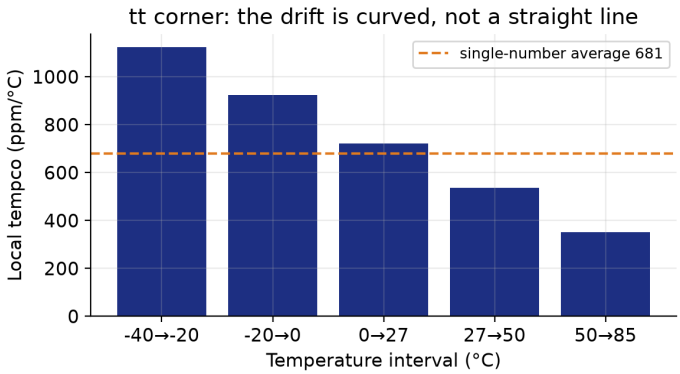


How to read this figure

Axes: left: x temperature (°C), y frequency change relative to 27 °C (%), at fixed $V_{ctrl} = 0.9$ V, one line per PDK corner; the green band is our ±0.3% spec. Right: bar chart of the average tempco in ppm/°C.
Good vs bad: Good would be a flat line inside the green band. Every corner is outside it except that sf only just stays near it. Steeper line = worse.
Key number: Tempco (PDK corners): sf 112, ff 332, tt 681, ss 1070, fs 1399 ppm/°C, a 12.5× spread. Typical-corner span 8.51%.
Takeaway: Frequency *rises* with temperature at this bias, and the amount depends strongly on the process corner, so no single fixed correction suits every chip.

Tempco $= (f_{85} - f_{-40})/(f_{27} \cdot 125)$. Span is max − min of f over the range, as a % of f_{27}. Procedure: fix $V_{ctrl} = 0.9$ V, simulate $T \in \{-40, -20, 0, 27, 50, 85\}$ °C.

3.1 Why the drift is curved



How to read this figure

Axes: x: temperature interval; y: local tempco (ppm/°C) over that interval at tt; dashed line is the single-number average.
Good vs bad: A perfectly linear drift would give equal bars. Ours fall from 1123 (cold) to 352 (hot).
Key number: Average 681 ppm/°C, cold end 1123, hot end 352.
Takeaway: The drift is curved, so a straight-line (1st-order) correction leaves a bowed residual, which motivates the $(T - 27)^2$ term.

4 Corner audit: why sf/fs numbers looked out of range

We were asked in review why ff/tt/fs/sf values "aren't in the ranges they're meant to be". We audited everything. **Nothing in the simulations was broken.** What was wrong was how we labelled and expected the corners.

4.1 Process corners from scratch

A MOSFET's threshold voltage V_{th}, mobility and so drive current vary from die to die and wafer to wafer. A *process corner* is a set of device models with these parameters pushed to an extreme. Two letters: the first for the NMOS, the second for the PMOS. **t** = typical, **s** = slow (higher V_{th}, lower current), **f** = fast (lower V_{th}, higher current). So **tt** is typical both, **ss** both slow, **ff** both fast, and the *skew* corners **sf** and **fs** have one device slow and the other fast. In this PDK, measurement shows **sf = fast NMOS / slow PMOS** and **fs = slow NMOS / fast PMOS**. (The PDK only says "Slow-Fast corner" without naming devices, and SkyWater's doc tables are inconsistent, so we trust our measurements.) Rule: always say "fast-NMOS", never assume "sf means slow N".

4.2 Why NMOS drive sets this VCO

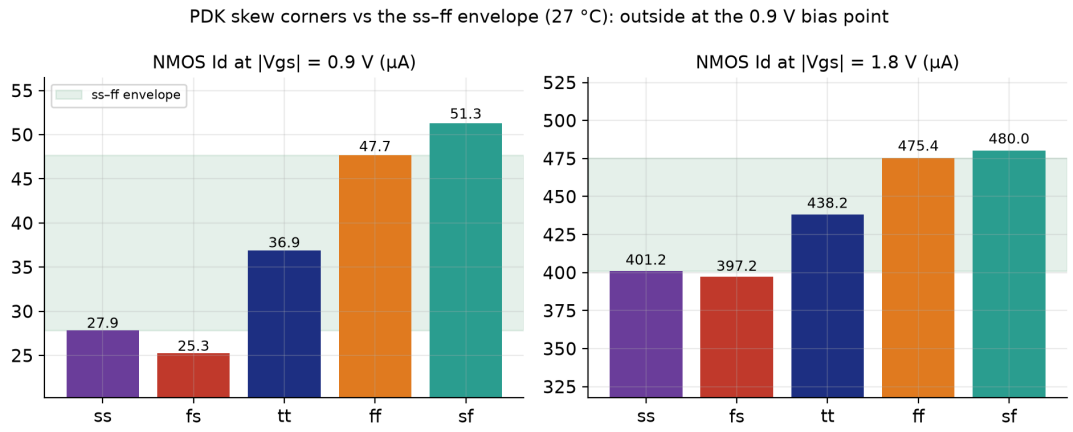
At $V_{ctrl} = 0.9$ V the NMOS starving transistor has an overdrive $V_{gs} - V_{th}$ of only about 0.2 V, since NMOS $V_{th} \approx 0.67$ V. The current through it sets the delay. The PMOS starving device is biased through a mirror and is the less limiting device at this bias. So the NMOS corner dominates: at 0.9 V going slow-to-fast NMOS changes f by about 36%, while slow-to-fast PMOS changes it by only about 10%. (At $V_{ctrl} = 1.4$ V this flips: PMOS 26% vs NMOS 9%.)

4.3 The PDK skew corners are wider than ss/ff

Extracted threshold shifts versus tt:

	ss	ff	sf	fs
NMOS V_{th} shift (mV)	+25	−25	−36	+36
PMOS $\lvert V_{th}\rvert$ shift (mV)	+15	−23	+32	−43

Intuition: we expected skew corners to be milder than ss/ff, lying between them. In this PDK they push each device *further* than ss/ff do. With a small overdrive (0.2 V) an extra 11 mV of V_{th} is about $2 \cdot 11/230 \approx 9\%$ of the drive current. So at the VCO bias the skews fall outside the ss–ff envelope.

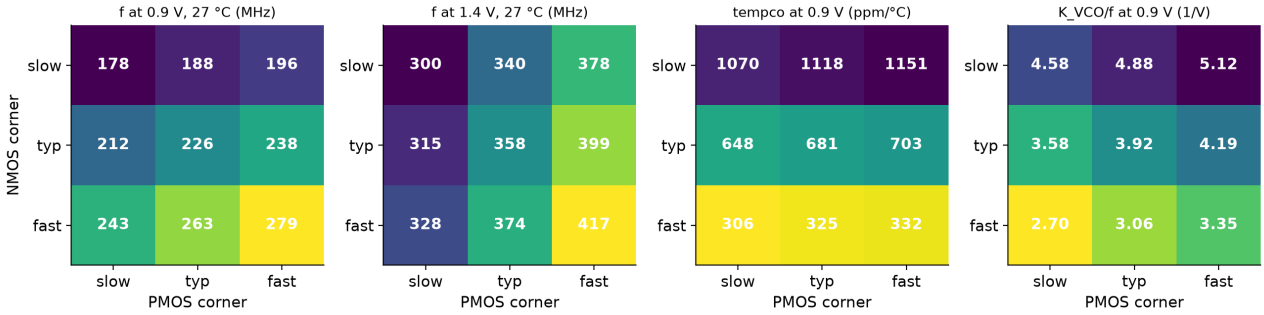


How to read this figure

Axes: two bar charts of NMOS drain current of a single 2/0.5 μm device at 27 °C: left at $|V_{gs}| = 0.9$ V (μA), right at 1.8 V; the green band is the ss–ff range.
Good vs bad: Textbook behaviour has every bar inside the band. On the left sf (51.3) and fs (25.3) stick out by 8–9% beyond ff (47.7) and ss (27.9); on the right (strong inversion) all are within 1%.
Key number: 0.9 V: fs 25.3, ss 27.9, tt 36.9, ff 47.7, sf 51.3 μA. 1.8 V: ss 401, fs 397, tt 438, ff 475, sf 480 μA.
Takeaway: The skew corners are not weird; at low overdrive the PDK's wider skew definition is amplified. Our VCO happens to sit at low overdrive.

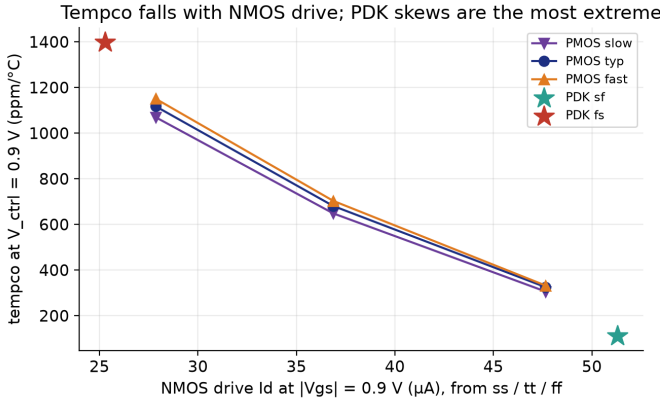
4.4 The textbook 3×3 corner set

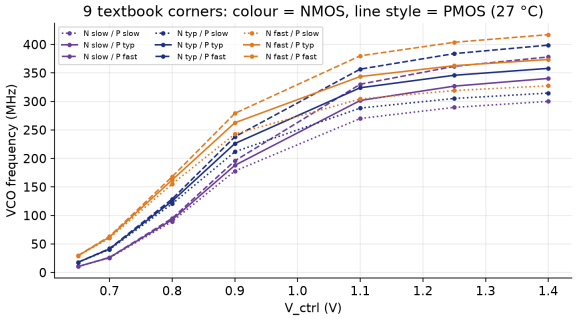
To give a clean reference we built all nine NMOS × PMOS combinations (slow/typ/fast each) by swapping the PDK's own NMOS and PMOS model files independently (`scripts/make_factorial_lib.py`). The diagonal reproduces the PDK ss/tt/ff exactly (177.993/226.206/279.178 MHz), a good validity check. Naming: `nXpY` (X,Y in s/t/f); `nsps`=ss, `ntpt`=tt, `nfpf`=ff.



How to read this figure

Axes: four heatmaps; rows are the NMOS corner (slow, typ, fast), columns the PMOS corner. Panels: f at 0.9 V; f at 1.4 V; tempco at 0.9 V (ppm/°C); K_{VCO}/f at 0.9 V (1/V). Colour is only relative within a panel.
Good vs bad: A textbook-behaving set is smooth and monotonic: f increases down the rows and right across the columns; tempco *decreases* with NMOS strength. All three hold (and the checks pass automatically: 8 monotonicity checks).
Key number: f(0.9 V): N slow 178.0/188.0/195.9; N typ 211.8/226.2/238.1; N fast 242.8/262.7/279.2 MHz. Tempco: 1070–1151 (N slow), 648–703 (N typ), 306–332 (N fast).
Takeaway: With independent N and P, the VCO behaves exactly as device physics predicts, and NMOS strength dominates at 0.9 V. Tempco range: textbook 306–1151 (3.8×); PDK sf/fs 112–1399 (12.5×).





How to read this figure

Axes: left: x NMOS drive current at 0.9 V (μA, from the ss/tt/ff devices), y tempco (ppm/°C); one line per PMOS state; stars are the PDK sf/fs corners. Right: tuning curves for the 9 textbook corners (colour = NMOS, line style = PMOS).
Good vs bad: Left: a straight falling line means tempco is a function of NMOS drive, which is the case. The stars lie on the extension of that trend, further out than ss/ff. Right: the colour groups (NMOS) separate more than the line styles (PMOS) at 0.9 V; they cross over at higher V_{ctrl}.
Key number: Tempco 1070 at 28 μA down to about 325 at 47.7 μA; PDK fs 1399 at 25.3 μA, PDK sf 112 at 51.3 μA.
Takeaway: The star points confirm the extremes are explained by NMOS drive, not by a bug.

4.5 Physics sanity checks (all pass)

- V_{th} falls with temperature: −0.78 mV/°C (NMOS), −1.1 mV/°C (PMOS); textbook range −0.5 to −3.
- Drain current at 1.8 V *falls* with temperature (−30% over 125 °C, mobility dominates); at 0.9 V it *rises* (+14%, V_{th} drop dominates). So the 0.9 V bias is below the zero-tempco point, which is why f rises with T. At $V_{ctrl} = 1.4$ V (more overdrive) the tempco is −816 ppm/°C: the sign flips.
- Supply current is proportional to frequency ($C \approx 27$–30 fF per node) in every corner.
- All ordering checks pass, except that the PDK sf/fs *tempcos* are outside [ff, ss] = [332, 1070] (expected, see above); their *frequencies* are inside the ss–ff envelope at 0.9 and 1.4 V.
- Closed-loop lock voltage equals the open-loop voltage for 226.2 MHz within 0.2 mV at all 11 corners (Section 8).

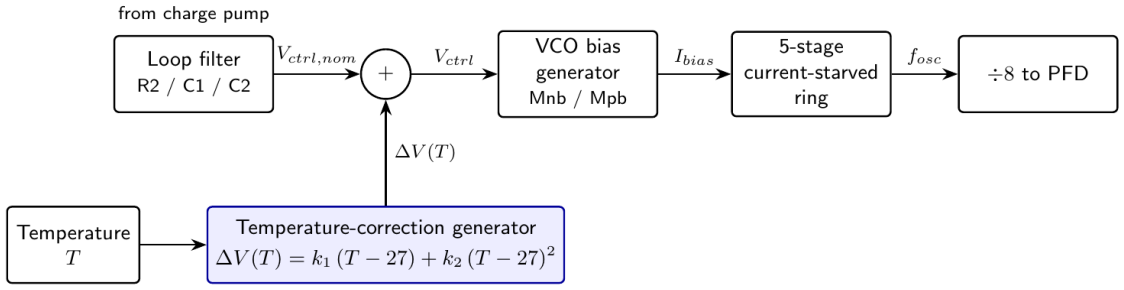
Be careful

Two PDK model sets exist (legacy per-corner files, which our ngspice library uses, and "continuous" parameter files where the skew shifts are ±27 mV vs ±15 mV for ss/ff). The qualitative story is the same. We cannot claim to know the foundry's intent; we report measured behaviour.

5 VCO characterisation and the tempco mechanism

- **Grid:** 5 PDK corners × 6 temperatures ($-40, -20, 0, 27, 50, 85$ °C) at $V_{ctrl} = 0.9$ V, plus the 3×3 textbook set.
- **Mechanism (supported interpretation):** V_{ctrl} drives the NMOS starving gate; at 0.9 V its overdrive is small. When temperature rises, V_{th} drops by ∼0.8 mV/°C (more current) and mobility drops (less current). At low overdrive the V_{th} effect wins, so current and frequency rise. Less overdrive ⇒ stronger positive tempco; this is exactly the corner ranking. At 1.4 V mobility wins and the tempco is negative. A separate isolation experiment was not run, so we call this a well-supported interpretation, not a proof.
- **Why "NMOS drive" and not "N/P balance":** the earlier explanation (mismatched corners balancing the current sources) was wrong; see Section 10.

6 The compensation network and how the fit works



1st-order: $k_2 = 0$ (linear, PTAT/CTAT-equivalent) **2nd-order:** $k_2 \neq 0$ (curvature-matched)
Coefficients fitted per process corner; an ideal behavioural source in Review 2.

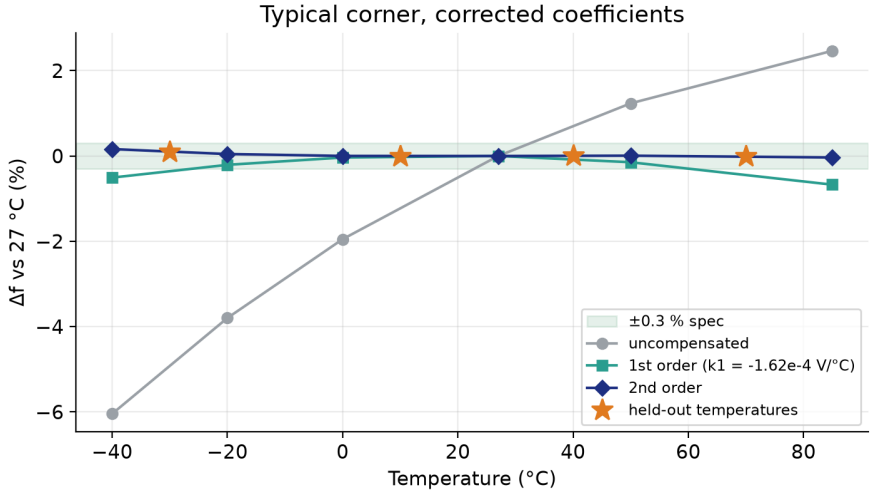
$$V_{ctrl,eff} = V_{ctrl} + k_1(T-27) + k_2(T-27)^2$$

- **Implementation:** an ideal behavioural voltage source (ngspice B-source reading the simulator temperature). It is *not* transistors; in silicon a PTAT/CTAT current generator plus a squarer would do this. The compensation is nominal (no mismatch).
- **How the new fit works** (`scripts/comp_factorial.py`): (1) measure $f(T, V)$ at $V = 0.85/0.9/0.95$ V for 6 temperatures; (2) at each temperature solve for the exact voltage shift $\Delta V^*(T)$ that restores $f(27\,°\text{C})$; (3) least-squares fit k_1 (1st order: $k_1(T-27)$ only) and k_1, k_2 (2nd order); (4) *verify* by simulating the compensated VCO at the 6 fit temperatures and at 4 held-out temperatures. This uses the true local sensitivity of the circuit, not a guess for K_{VCO}.
- **tt coefficients:** 1st-order-only fit $k_1 = -1.623 \times 10^{-4}$ V/°C; 2nd-order fit $k_1 = -1.543 \times 10^{-4}$ V/°C, $k_2 = 4.18 \times 10^{-7}$ V/°C^2. Because frequency rises with T, $k_1 < 0$ (it lowers the control voltage when hot).

Definitions.

- **Span:** max − min of f over −40 to 85 °C, as % of f_{27}.
- **Max deviation:** largest $|f - f_{27}|/f_{27}$ over the range, in %. The spec ±0.3% is judged on this.
- **Held-out:** temperatures $(-30, 10, 40, 70\,°\text{C})$ never used in the fit, simulated afterwards as an overfitting guard (2 coefficients, 5 independent temperature points).
- **Compensation ratio** C**:** baseline span / 2nd-order span. Bigger is better.

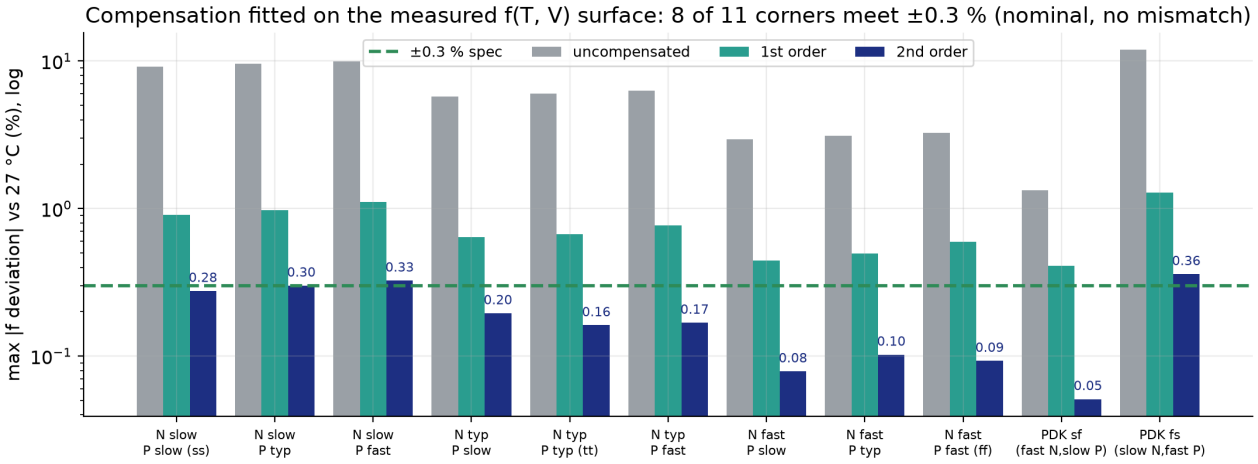
6.1 Typical corner



How to read this figure

Axes: x: temperature (°C); y: frequency change relative to 27 °C (%). Grey: uncompensated; teal squares: 1st order; navy diamonds: 2nd order; orange stars: the held-out temperatures (second-order result). Green band: ±0.3% spec. The legend's $k_1 = -1.62 \times 10^{-4}$ belongs to the 1st-order-only fit.
Good vs bad: Good = the curve stays inside the green band. Grey leaves it (−6% to +2.5%); 1st order bows out to about −0.65% at 85 °C; 2nd order stays in the band and the stars (never fitted) sit on the same line.
Key number: Span 8.51% → 0.67% (1st) → 0.20% (2nd); max deviation 0.16% (spec met); held-out max dev 0.09%; $C = 43$.
Takeaway: With correctly fitted coefficients the 2nd-order network meets spec at tt, and the held-out points show it is not overfitted.

6.2 All 11 corners



How to read this figure

Axes: x: the 11 corners (textbook nXpY, with ss/tt/ff marked, plus PDK sf and fs); y: max $|f - f_{27}|$ in % on a *log* axis. Grey: uncompensated; teal: 1st order; navy: 2nd order (value printed). The dashed green line is the ±0.3% spec.
Good vs bad: A navy bar below the dashed line passes. Each log gridline is a factor of 10, so the grey-to-navy gap of 1–2 decades is a 30–50× improvement. Bars above the line are failures.
Key number: 8 of 11 meet spec; the misses are N slow/P typ 0.301, N slow/P fast 0.325 and PDK fs 0.360.
Takeaway: Every corner improves by one to two orders of magnitude; the three misses are all slow-NMOS corners, where the drift is largest and the fit has the most to correct.

Corner	Base %	1st %	2nd %	C	max dev %	held-out %	Meets?
ss (nsps)	13.37	0.91	0.35	38.4	0.277	0.175	yes
N slow/P typ	13.97	0.98	0.38	37.2	0.301	0.201	no
N slow/P fast	14.38	1.10	0.42	34.3	0.325	0.223	no
N typ/P slow	8.10	0.64	0.24	33.6	0.195	0.109	yes
tt (ntpt)	8.51	0.67	0.20	43.2	0.162	0.091	yes
N typ/P fast	8.79	0.77	0.20	43.7	0.168	0.097	yes
N fast/P slow	3.82	0.45	0.12	31.0	0.079	0.034	yes
N fast/P typ	4.06	0.50	0.12	33.3	0.102	0.041	yes
ff (nfpf)	4.15	0.59	0.11	37.0	0.093	0.035	yes
PDK sf (fast N/slow P)	1.49	0.41	0.09	15.9	0.051	0.036	yes
PDK fs (slow N/fast P)	17.49	1.29	0.49	35.8	0.360	0.251	no

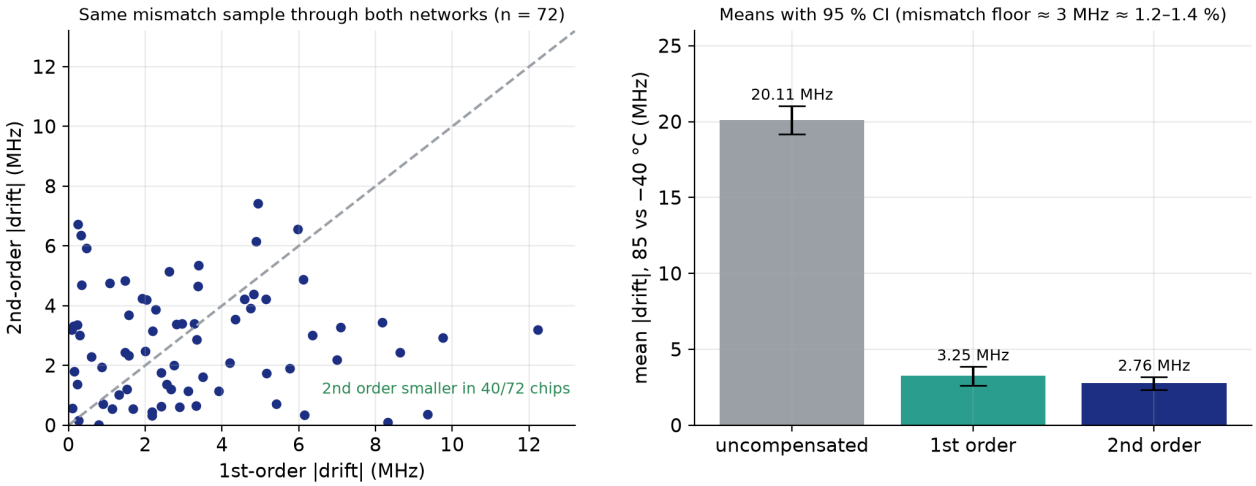
7 of the 9 textbook corners meet spec; 8 of all 11. C ranges 31–44 except PDK sf (15.9), whose baseline is already tiny. Held-out max deviation ranges 0.03–0.25%.

Which do not meet, and why. N slow/P typ (0.301, borderline: it misses by 0.001 points), N slow/P fast (0.325) and PDK fs (0.360). Slow NMOS means the smallest overdrive, the largest and most curved drift (baseline spans 14–17.5%), and a residual that a single quadratic cannot flatten to better than about 0.3–0.36%. They are close; we do not call them "met".

Be careful

Caveats to state yourself: the source is ideal; this is nominal (no mismatch); the fit is at $V_{ctrl} = 0.9$ V; 2 coefficients are fitted to 5 temperature points (mitigated by the held-out check). Also, each corner locks at its own V_{ctrl} (0.85–0.97 V), so the ideal target would be the compensation at that corner's own lock voltage (future work).

7 Monte Carlo, $n = 72$



How to read this figure

Axes: Left: each dot is one virtual chip (one random mismatch seed at tt). x: its |drift| = $|f(85) - f(-40)|$ with the 1st-order network (MHz); y: the same chip with the 2nd-order network. Dashed line: equal. Right: bar chart of the mean |drift| with 95% confidence intervals.
Good vs bad: Left: dots *below* the dashed line are chips where 2nd order is better. A clear win would put almost all dots below it; here they are scattered on both sides (40 of 72 below). Right: a good result is a low bar with a small interval; the compensated bars are far below the grey one but overlap each other.
Key number: Mean |drift|: 20.11 MHz [19.18, 21.04] uncompensated; 3.25 MHz [2.63, 3.87] 1st order; 2.76 MHz [2.33, 3.19] 2nd order.
Takeaway: Compensation works on mismatched chips (72/72 improve), but 2nd order is not distinguishable from 1st order. The remaining 3 MHz is a mismatch floor.

7.1 What a paired test is, in plain language

Each seed is a virtual chip with its own random transistor mismatch. Chips differ a lot from one another, which would drown a small difference between two networks. *Pairing* means we run the *same* chip through both networks and look only at the per-chip difference, so chip-to-chip variation cancels. We then ask: is the mean of those differences distinguishable from zero? The 95% confidence interval (CI) is the range likely to contain the true mean difference; if it contains 0, the effect is not significant. The paired t-test assumes roughly normal differences; Wilcoxon signed-rank uses ranks only and does not. We report both; they agree. The metric is |drift| (not signed drift; see Section 10).

7.2 Results

Comparison	Mean reduction (MHz)	95% CI	Significant?
Baseline → 1st order	16.86	[15.57, 18.16]	yes (t and Wilcoxon), better in 71/72
Baseline → 2nd order	17.35	[16.37, 18.34]	yes (both), better in 72/72
1st → 2nd order	0.49	[−0.26, 1.24]	**no** (both), 2nd smaller in 40/72

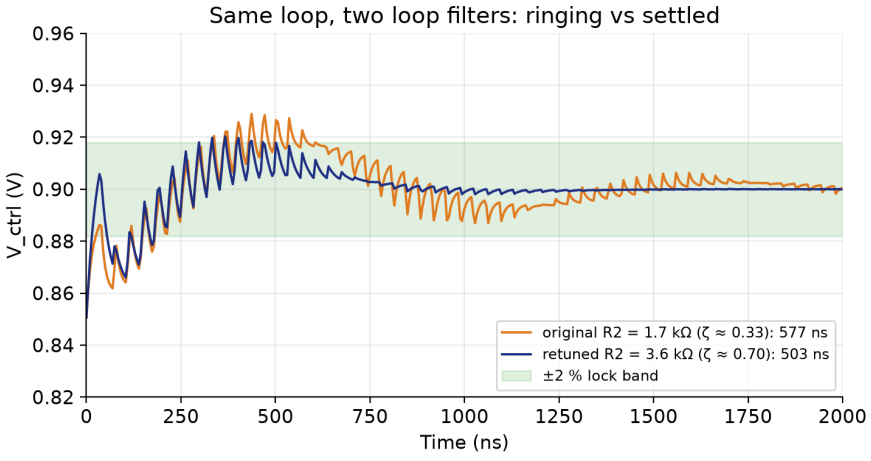
7.3 The mismatch floor, and why this is an honest finding

Nominally the residual is only 0.2–0.7%. With mismatch each chip has its own drift curve, but we apply one global correction fitted to the nominal chip. What cannot be removed is each chip's own deviation: about 3 MHz, which is 1.2–1.4% of 226 MHz. Signed-drift standard deviation is 4.1 MHz (1st) and 3.3 MHz (2nd): the spread across chips, not the average, is the problem. The 2nd-order term refines the *average* shape; but the error per chip is dominated by mismatch, so improving the mean shape by 0.2–0.5% does not show up. The CI for 1st vs 2nd order includes zero, so we cannot claim 2nd order helps under mismatch. This is the correct conclusion with correctly fitted coefficients. Getting to ±0.3% across chips would need a per-chip trim (future work).

8 Closed loop

8.1 Loop dynamics, found, fixed, verified

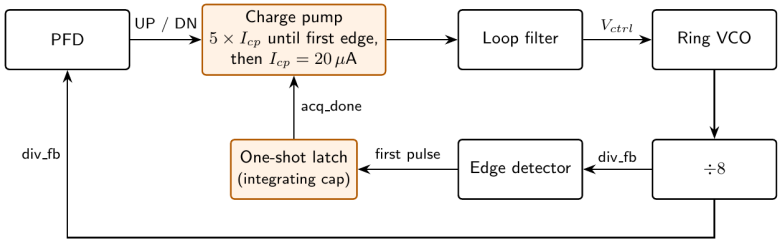
- The loop is a second-order system with natural frequency $\omega_n^2 = I_{cp}K_{VCO,rad}/(2\pi NC_1)$ and damping $\zeta = R_2C_1\omega_n/2$. The $1/(2\pi)$ comes from the PFD/charge-pump gain ($I_{cp}/2\pi$ amps per radian of phase error). Review 1 dropped it.
- With the measured $K_{VCO} = 887$ MHz/V and the original $R_2 = 1.7$ kΩ: $f_n \approx 0.91$ MHz, $\zeta \approx 0.33$ (underdamped, rings). Review 1 had claimed $\zeta = 0.72$.
- ζ depends on R_2 at fixed f_n, so we raised R_2 to 3.6 kΩ: $f_n \approx 0.91$ MHz, $\zeta \approx 0.70$, phase margin about 65–70 *by the* 100ζ *rule of thumb*, not a Bode plot.



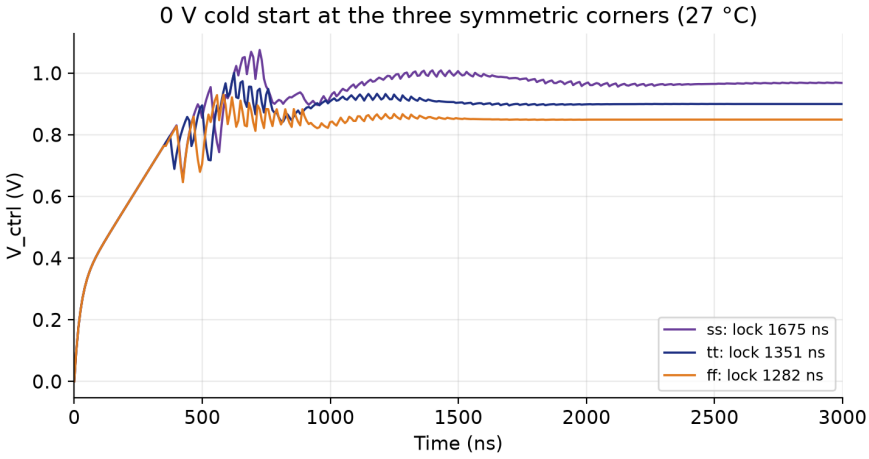
How to read this figure

Axes: x: time (ns); y: V_{ctrl} (V), tt, 27 °C, coarse start at 0.85 V. Orange: original filter; navy: retuned. Green band: ±2% of the final 0.9 V, the lock definition.
Good vs bad: Good = enter the band and stay, with small ripple. Orange keeps ringing up to 2 μs; navy settles smoothly. The sawtooth is the per-cycle charge-pump pulses.
Key number: Lock 577 → 503 ns; ripple 8.6 → 0.35 mV; final frequency error 0.78% → 0.008%; peak V_{ctrl} 0.930 → 0.922 V.
Takeaway: The retune removed the ringing; only $R_2 = 1.7$, 3.0 and 3.6 kΩ were simulated at transistor level, so 3.6 is not claimed optimal.

8.2 Cold start



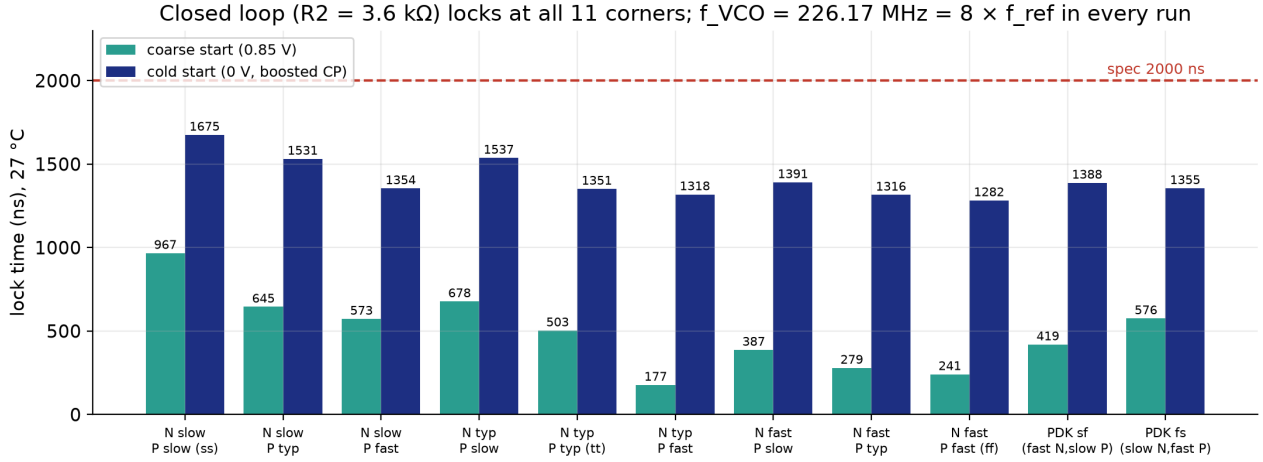
At 0 V the VCO is nearly off, so the divided clock is absent and the nominal 20 μA pump is too slow. The fix: run the charge pump at $5 \times I_{cp}$ until the divider produces its first edge, then drop to 20 μA. This is done by an edge detector feeding a one-shot latch. The price of the retune is that the cold start became slower (839 ns with the original filter → 1351 ns at tt); both are inside 2 μs.



How to read this figure

Axes: x: time (ns); y: V_{ctrl} (V); true 0 V start at 27 °C for ss (purple), tt (navy), ff (orange).
Good vs bad: Good = a fast monotonic climb, a limited overshoot, and a flat settled value. The three curves rise together for the first 350 ns (the boosted charge pump), then ring while the loop captures phase, then settle at different final voltages.
Key number: Lock 1675 ns (ss), 1351 ns (tt), 1282 ns (ff); settled V_{ctrl} 0.968, 0.900, 0.849 V.
Takeaway: The boost drives V_{ctrl} through the dead region quickly; the loop then locks inside spec at every corner shown. ss is slowest because it needs the highest V_{ctrl}.

8.3 Lock across 11 corners

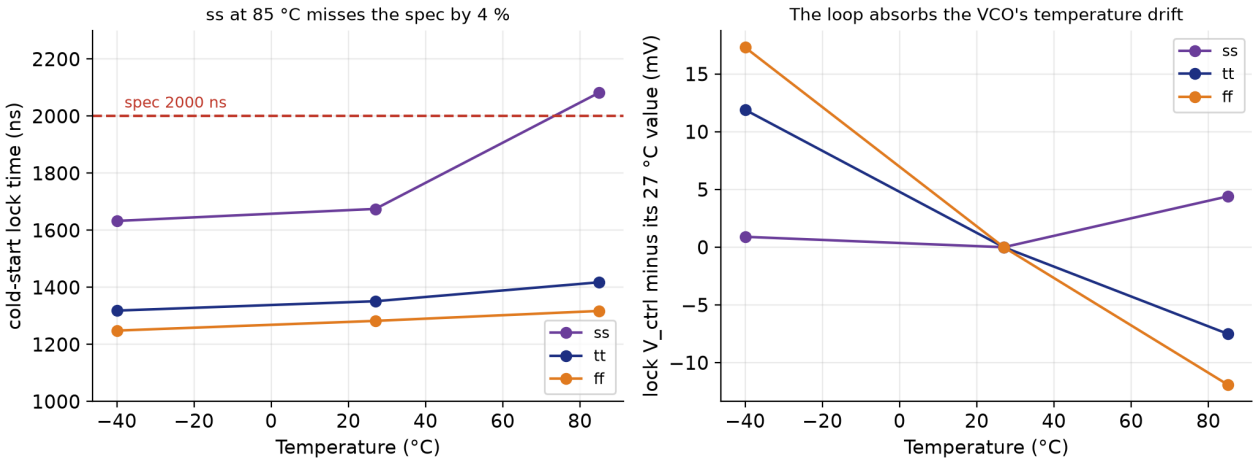


How to read this figure

Axes: x: the 11 corners; y: lock time (ns) at 27 °C. Teal: coarse start (0.85 V). Navy: cold start from 0 V with boosted charge pump. Red dashed line: 2000 ns spec.
Good vs bad: A bar below the red line passes; they all do at 27 °C. The title notes the output frequency: 226.17 MHz = $8 \times f_{ref}$ in every run.
Key number: Coarse: 177–967 ns (ss slowest 967). Cold: 1282–1675 ns. Ripple ≤ 0.8 mV, all runs settled, $f_{VCO} = 226.17$ MHz (-0.01%, the crossing-detector bias).
Takeaway: The same loop locks across all process corners, with comfortable margin at 27 °C.

Corner	Coarse (ns)	Cold (ns)	Lock V_{ctrl} (V)
ss (nsps)	967	1675	0.968
N slow/P typ	645	1531	0.944
N slow/P fast	573	1354	0.930
N typ/P slow	678	1537	0.920
tt (ntpt)	503	1351	0.900
N typ/P fast	177	1318	0.888
N fast/P slow	387	1391	0.877
N fast/P typ	279	1316	0.860
ff (nfpf)	241	1283	0.849
PDK sf	419	1388	0.869
PDK fs	576	1355	0.942

8.4 Temperature, and the ss/85 °C miss

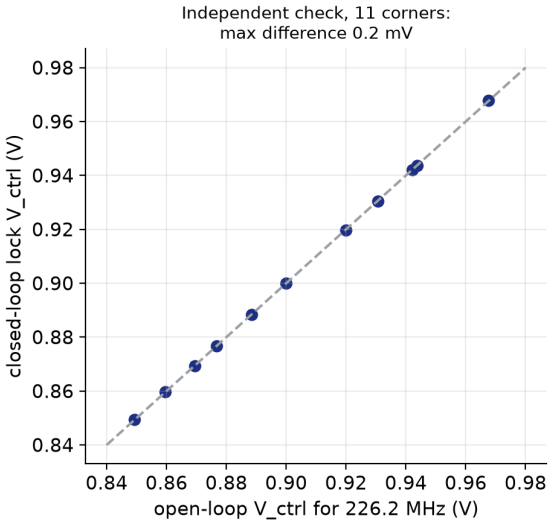


How to read this figure

Axes: Left: x temperature (°C); y cold-start lock time (ns) for ss, tt, ff; red dashed = 2000 ns spec. Right: x temperature; y change in the locked V_{ctrl} relative to its 27 °C value (mV).
Good vs bad: Left: any point above the red line fails. Right: the loop absorbs the VCO's drift by moving V_{ctrl}, and a larger slope means more drift absorbed.
Key number: ss: -40 °C 1633 ns, 27 °C 1675, **85 °C 2083 ns (misses 2 μs by 4%)**; tt 1319/1351/1418; ff 1249/1283/1317. Locked V_{ctrl} at tt: +11.9 mV at -40 °C, -7.5 mV at 85 °C.
Takeaway: The worst case, ss at 85 °C, misses the lock-time spec slightly. We report it as is; the lock frequency itself is still correct (226.11 MHz).

In lock the PLL absorbs the VCO's tempco. At tt the lock-voltage shift (+11.9/−7.5 mV) matches the fitted compensation correction (+12.2/−7.5 mV): an independent confirmation of the compensation fit *at tt*. At other corners this comparison is not valid, because the compensation was fitted at 0.9 V while each corner locks at its own voltage. Temperature runs: ss, tt and ff only, cold start; not all 11 corners at all temperatures.

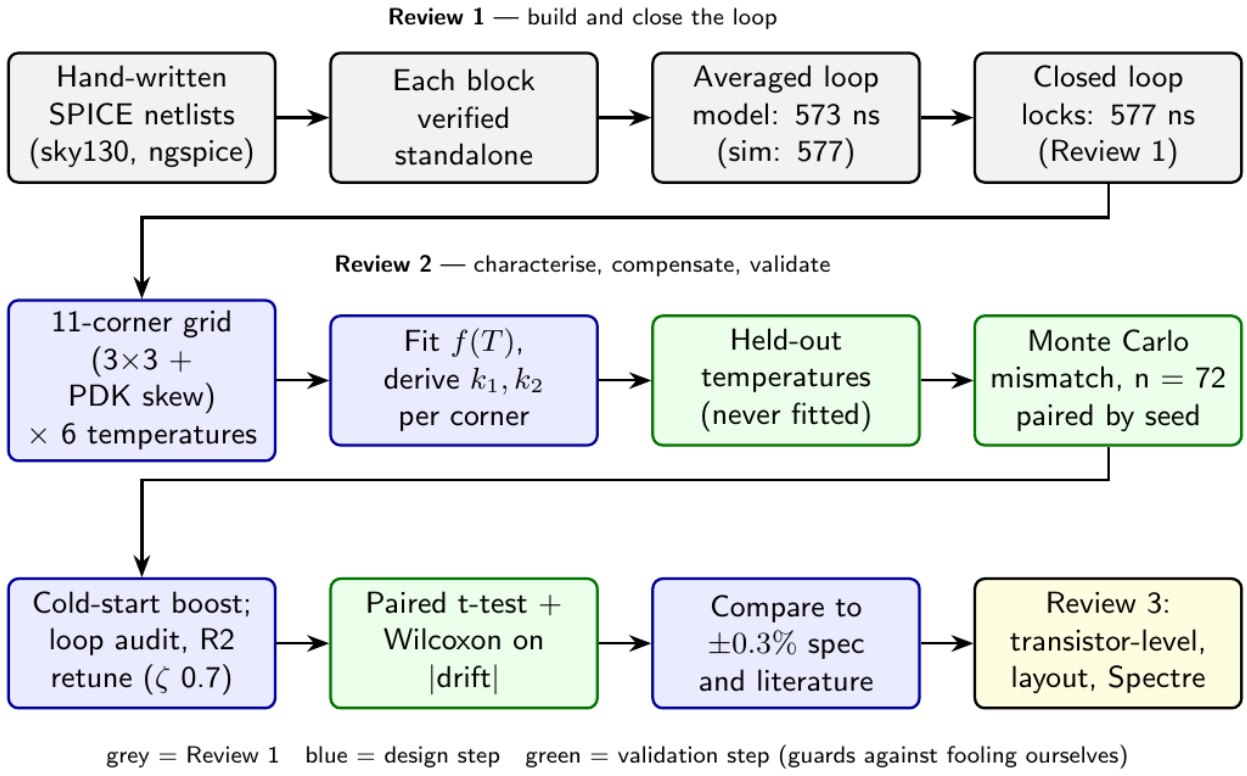
8.5 Closed-loop vs open-loop voltage check



How to read this figure

Axes: x: the open-loop V_{ctrl} that gives 226.2 MHz (from the free-running VCO); y: the locked V_{ctrl} in the closed loop; one dot per corner; dashed 45 line = perfect agreement.
Good vs bad: Good = every point on the dashed line. All 11 do.
Key number: Maximum difference 0.2 mV over 0.849–0.968 V.
Takeaway: Two independent simulations (open-loop VCO and closed loop) agree, so the loop is locking where the VCO actually produces 226.2 MHz.

9 Method, novelty and limitations



This is the Review 1/2/3 flow diagram; its Monte Carlo box ("$n = 30$") is the Review 2 size. The final analysis uses $n = 72$.

- **Method:** simulate everything at transistor level in ngspice 47 with sky130B (volare commit c6d73a35); characterise → model/fit → verify on held-out data → test statistically. The green steps (held-out, paired statistics, independent checks) are there to stop us fooling ourselves.
- **Literature and novelty:** US Patent 9,748,962 B2 (Infineon, 2017) already covers generic quadratic temperature compensation. We do *not* claim quadratic compensation as new. Our claim is narrower: a closed-loop, per-corner, statistically validated study of it for a ring-VCO PLL on an open PDK. Benchmarks: Souliotis 2022, Stancu 2023, Geng 2018 (`docs/benchmark_table.md`); compare honestly, since we simulate an ideal network and they measure silicon.
- **Hypothesis H1** (residual error grows like $C\cdot$ raw error) needs Cadence Virtuoso/Spectre access, still pending. Our α sweep is near-circular and is not the H1 test.

Limitations, stated plainly.

- Compensation and charge pump are behavioural (ideal); no transistor-level versions yet.
- No layout, DRC, LVS or post-layout simulation.
- No jitter or phase-noise number (no PSS/Pnoise in open tools).
- Phase margin is the 100ζ rule, not a measured Bode plot.
- Only $R_2 = 1.7/3.0/3.6$ kΩ simulated; 3.6 not claimed optimal.
- Temperature runs of the closed loop cover 3 corners only; the cold start across corners is at 27 °C.
- Compensation is nominal and fitted at $V_{ctrl} = 0.9$ V; mismatch limits compensated drift to about 1.2–1.4%.
- ss at 85 °C cold start (2083 ns) exceeds the 2 μs spec.

Still open: transistor-level charge pump and compensation network (PTAT/CTAT + squarer) and a loop re-run; per-chip trim; compensation at each corner's own lock voltage; layout in Magic, DRC, LVS with netgen, post-layout simulation; Spectre arm, PSS/Pnoise, H1 regression (confirm lab access with Dr. Bindu); spec sign-offs; paper and artifact release.

10 Corrections log

Corrections (own these)

After Review 2 we re-checked our own work. These nine items were wrong or changed; say so openly. It strengthens credibility.

1. **sf/fs labels reversed.** sf is fast-NMOS/slow-PMOS, fs is slow-NMOS/fast-PMOS.
2. **Tempco mechanism.** It follows NMOS overdrive, not an N/P balance; the sign flips at 1.4 V (−816 ppm/°C).
3. **K_{VCO}:** 669 MHz/V (a fit over the whole S-curve) was wrong; the local value at 0.9 V is about 890 (887).
4. **Oscillation floor.** The "0.63 V hard floor" was a short-window artefact; the ring oscillates at low frequency (0.63 MHz at 0.5 V).
5. **Loop damping.** Review 1 dropped the $1/2\pi$, so ζ was 0.33, not 0.72; R_2 retuned 1.7 → 3.6 kΩ, $\zeta \approx 0.70$, and verified.
6. **Compensation coefficients.** The Review 2 coefficients were mis-estimated because of the wrong K_{VCO} and an analytic 1st-order guess. Review 2 reported tt span 1.12%, max dev 0.87%, spec missed, and "met at ff and sf only (2 of 5)", C of 3.3–40×. The corrected fit gives tt 0.20% span, 0.16% max deviation (spec met), and 8 of 11 corners.
7. **Monte Carlo.** Review 2 claimed 2nd order significantly better than 1st (−2.48 MHz, CI [0.81, 4.15], 23/30 seeds). That compared against a badly mis-tuned 1st-order arm ($k_1 = -2.163 \times 10^{-4}$ from the wrong K_{VCO}). With correct coefficients and $n = 72$: 0.49 MHz, CI [−0.26, 1.24], not significant. (Earlier still, the first analysis tested *signed* drift, which was wrong; it was corrected to |drift|.)
8. **Skew corners.** We had expected the PDK skew corners to lie inside the ss–ff envelope; they do not (wider shifts). We added the textbook 3×3 set and report 11 corners.
9. **ss/85 °C.** The closed-loop campaign found the cold start misses 2 μs by 4% (2083 ns).

Unchanged and still valid: PDK corner tempcos 112/332/681/1070/1399 ppm/°C; f at 0.9 V per corner; the tuning range; the uncompensated spans; lock times of the retuned loop at the five original corners (they were re-run in the campaign and match).

11 Numbers to memorise

Reference / output / N	28.275 MHz / 226.2 MHz at 0.9 V / 8
Loop filter, I_{cp}	R_2 3.6 kΩ (was 1.7), C_1 68 pF, C_2 6.8 pF; 20 μA (5× cold boost)
Loop	$f_n \approx 0.91$ MHz; ζ 0.33 → 0.70; PM ≈65–70 (100ζ rule)
K_{VCO} tt, 0.9 V	≈890 MHz/V (887); sf/ff/ss/fs 580/940/820/1020
PDK tempcos sf/ff/tt/ss/fs	112/332/681/1070/1399 ppm/°C (12.5×); textbook 306–1151 (3.8×)
f at 0.9 V, ss/fs/tt/sf/ff	178.0 / 183.7 / 226.2 / 246.2 / 279.2 MHz
Vth tempco	−0.78 (N), −1.1 (P) mV/°C; tempco at 1.4 V: −816 ppm/°C
Skew-corner ΔV_{th} (N)	ss +25, ff −25, sf −36, fs +36 mV
tt span: base → 1st → 2nd	8.51 → 0.67 → 0.20%; max dev 0.16%; held-out 0.09%; $C = 43$
tt coefficients	$k_1 = -1.543 \times 10^{-4}$ V/°C, $k_2 = 4.18 \times 10^{-7}$ V/°C^2
Spec ±0.3% met	8 of 11 corners; misses 0.301, 0.325, 0.360%
Monte Carlo ($n = 72$, tt)	20.11 / 3.25 / 2.76 MHz; 1st vs 2nd 0.49 MHz, CI [−0.26, 1.24], not significant; floor ≈3 MHz (1.2–1.4%)
Lock tt, coarse / cold	503 ns / 1351 ns (spec <2000)
Lock range, 27 °C	coarse 177–967 ns; cold 1282–1675 ns (11 corners)
Worst case	ss, 85 °C, cold: 2083 ns (4% over)
f_{VCO} in lock	226.17 MHz = $8 \times f_{ref}$ (−0.01%); ripple ≤ 0.8 mV
Lock V_{ctrl} range	0.849 (ff) to 0.968 V (ss); open- vs closed-loop agree within 0.2 mV
Ripple / error (tt, retune)	8.6 → 0.35 mV; 0.78 → 0.008%
Temperatures	fit $-40, -20, 0, 27, 50, 85$; held-out $-30, 10, 40, 70$ °C

12 Likely examiner questions and model answers

Q: Why do the sf and fs corners look out of range?

They are not broken. In this PDK the skew corners shift each device's threshold by more than ss/ff do (NMOS V_{th}: sf −36, fs +36 mV against ∓25 for ff/ss). Our VCO runs at only 0.2 V overdrive, which amplifies an extra 11 mV into about 9% of drive current, so at 0.9 V the skews lie outside the ss–ff envelope (NMOS current: fs 25.3, ss 27.9, ff 47.7, sf 51.3 μA). At 1.8 V, in strong inversion, they are within 1%.

Q: Are your corners right?

Yes, as far as we can verify. We measured device behaviour, not assumed it: sf is fast-NMOS/slow-PMOS. We reproduced ss/tt/ff exactly with an independent textbook set that swaps the PDK's own NMOS and PMOS models, and all ordering checks pass. We earlier had the labels reversed and corrected it. The PDK documentation does not clearly name the devices, so we rely on measurement.

Q: What are process corners?

Sets of device models at the edges of manufacturing variation. First letter NMOS, second PMOS, s/t/f for slow/typical/fast. Slow means higher threshold and lower current. ss and ff move both together; sf/fs move them oppositely.

Q: Why does NMOS drive set this VCO?

At 0.9 V the NMOS starving transistor has small overdrive and its current sets the stage delay. Frequency changes 36% from slow to fast NMOS but 10% from slow to fast PMOS. At 1.4 V the PMOS dominates instead.

Q: Why did compensation improve so much compared with Review 2?

Because the earlier coefficients were wrong, not because the circuit changed. They used a wrong K_{VCO} (669 instead of about 890 MHz/V) and an analytic guess. The new method measures f at three control voltages and six temperatures, solves exactly for the voltage that restores f, fits k_1, k_2 by least squares, and verifies by simulating the compensated VCO, including held-out temperatures. tt went from 1.12% span to 0.20%.

Q: How do you know you did not overfit?

Four held-out temperatures ($-30, 10, 40, 70$ °C), never used in the fit, are simulated at every corner. Held-out max deviation is 0.03–0.25%, in the same range as the fitted points. Also the tt lock-voltage shift in closed loop (+11.9/−7.5 mV) matches the fit (+12.2/−7.5 mV).

Q: Why is 2nd order not significantly better than 1st?

With correct coefficients the 1st-order network already removes most of the drift (tt: 8.51 → 0.67%), and under mismatch each chip's residual is set by its own transistor mismatch, not the shape of the average curve. The paired difference is 0.49 MHz with CI [−0.26, 1.24], which includes zero; 2nd order wins in only 40 of 72 chips. This is an honest result. Our earlier claim of significance was an artefact of a mis-tuned 1st-order arm.

Q: What is the mismatch floor?

About 3 MHz (1.2–1.4% of 226 MHz) of residual drift that remains after compensation across 72 mismatched chips. One global correction cannot cancel each chip's individual drift. Beating it needs a per-chip trim.

Q: What is a paired test and why use it?

We run the same virtual chip through both networks and test the per-chip differences, so chip-to-chip variation cancels. A paired t-test and a Wilcoxon signed-rank test are used; they agree.

Q: What is your phase margin?

About 65–70 from the 100ζ rule with $\zeta \approx 0.70$. This is an estimate, not a measured Bode plot. The original $R_2 = 1.7$ kΩ gave $\zeta = 0.33$ and about 30. Review 1 claimed 0.72 because it dropped the $1/2\pi$ in the charge-pump gain.

Q: Why did you retune R_2 and what did it cost?

The corrected formulas showed the original loop was underdamped. R_2 1.7 → 3.6 kΩ keeps f_n and raises ζ to 0.70. It removed the ringing (ripple 8.6 → 0.35 mV, frequency error 0.78 → 0.008%, coarse lock 577 → 503 ns), but the cold start slowed from 839 to 1351 ns at tt. Both are within spec.

Q: Does it lock at every corner?

Yes at 27 °C, from both a 0.85 V coarse start (177–967 ns) and a 0 V cold start with the boosted charge pump (1282–1675 ns), all inside 2 μs, with output 226.17 MHz.

Q: Why does ss at 85 °C miss?

ss has the weakest NMOS so it needs the highest control voltage (0.968 V at 27 °C, 0.972 V at 85 °C), starting from 0 V it has the longest climb, and the cold start takes 2083 ns, 4% over the 2 μs spec. The frequency still locks correctly. We report it as a miss and propose a stronger boost or a faster filter at that corner as future tuning.

Q: What is ideal in your model?

The charge pump (behavioural current source with a boost switch) and the temperature compensation (an ideal voltage source reading temperature). The VCO, PFD, divider and loop filter are transistor-level or real components in ngspice with the sky130B models. No layout parasitics.

Q: What is novel, given the Infineon patent?

Quadratic temperature compensation is not novel (US 9,748,962 B2). Our claim is a closed-loop, per-corner, statistically validated study of it on a ring-VCO PLL with an open PDK, with honest negative findings such as the mismatch floor.

Q: Why a ring VCO rather than an LC oscillator?

A ring is small, needs no inductor, tunes widely and suits an open digital-friendly PDK. Its weakness is exactly what we study: large temperature sensitivity, plus worse phase noise.

Q: Why ngspice and sky130?

Both are fully open, so the work is reproducible by anyone; the PDK is pinned by volare commit. The trade-off is less mature models, hence the planned Spectre cross-check. ngspice 47 reproduces earlier ngspice 42 numbers exactly.

Q: Why is there no jitter or phase-noise result?

Open tools lack PSS/Pnoise. That belongs to the Spectre arm; we make no claim about noise.

Q: Why does frequency rise with temperature?

At 0.9 V the NMOS overdrive is small, so the drop in V_{th} (−0.78 mV/°C) beats the loss of mobility and the current increases. At 1.4 V the tempco is −816 ppm/°C. This is a well-supported interpretation backed by the sign flip and the ordering, not a proof.

Q: How did you get K_{VCO}?

As the local slope at 0.9 V, a central difference at ±50 mV: 887 MHz/V at tt. The earlier 669 came from fitting the whole S-shaped curve and was wrong.

Q: Does the VCO stop at 0.63 V?

No. That was a short-window artefact. It keeps oscillating (0.63 MHz at 0.5 V), but below 0.65 V it is under 8% of the target, so a 0 V start has no useful divided clock, hence the boost.

Q: What does the cold-start boost do to stability?

It acts only before the first divider edge, while the loop is open. After that the pump is at the nominal 20 μA for which $\zeta = 0.70$ was designed.

Q: Are 8 of 11 corners a pass or a failure?

Neither word fits. 8 meet ±0.3%; three miss by 0.001–0.06 points (0.301, 0.325, 0.360%), all slow-NMOS. Without any compensation, none would be near spec (1.5–17.5% spans). Closing the gap would take a per-corner coefficient, an extra term, or compensation at each corner's lock voltage.

Q: The compensation is an ideal source. How real is this?

It is a behavioural model, standard at this stage, that says what correction voltage is needed. In silicon, PTAT/CTAT currents plus a squarer would produce k_1 and k_2. That is the next step.

Q: Why a latch-based PFD?

A resettable D-flip-flop PFD needs a clean asynchronous reset, which fought the latch's feedback in our hand-built cells. Edge detectors plus NOR SR-latches were reliable and verified standalone.

Q: Your comparison with published work is simulation versus silicon. Is that fair?

No, so we say so in the benchmark table. It only shows that the simulated numbers are in the same range, not that we have equal performance.

Q: What remains?

Transistor-level charge pump and compensation network, loop re-run, per-chip trim, compensation at each corner's lock voltage, layout (Magic), DRC, LVS (netgen), post-layout simulation, Spectre/PSS/Pnoise and H1 test (needs lab access), a paper and artifact release.

Q: What would you do differently?

Define the compensation target at each corner's own lock voltage from the start, and check the $1/(2\pi)$ and K_{VCO} against simulation before building on them.

13 Glossary

Process corner
A device-model set at a manufacturing extreme: first letter NMOS, second PMOS; s/t/f = slow/typical/fast.

Textbook 3×3 set
The nine NMOS × PMOS combinations built from the PDK's own models; `nXpY`.

PDK sf / fs
The foundry's skew corners; here sf = fast N/slow P, fs = slow N/fast P; wider than ss/ff.

Overdrive
$V_{gs} - V_{th}$; sets drive current. About 0.2 V at our bias.

Tempco
Fractional frequency change per °C, in ppm/°C.

Span / max deviation
Max − min frequency, and largest $|f - f_{27}|$, in % of f_{27}; spec uses max deviation (±0.3%).

Held-out
Data kept out of the fit and used only to test it.

Compensation ratio C
Baseline span / compensated span.

PTAT / CTAT
Quantities proportional / complementary to absolute temperature.

K_{VCO}
df/dV_{ctrl}; about 890 MHz/V at tt, 0.9 V.

Damping ζ, phase margin
$\zeta = R_2 C_1 \omega_n / 2$; PM $\approx 100\zeta$ degrees (rule of thumb).

Lock time
Time until V_{ctrl} stays within $\pm 2\%$ of its final value.

Cold start
Starting from $V_{ctrl} = 0$ V with the boosted charge pump until the first divider edge.

Seed / Monte Carlo
One random mismatch instance (a virtual chip); $n = 72$ here.

Paired t-test, Wilcoxon
Tests of per-chip differences; t assumes normality, Wilcoxon uses ranks.

95% CI
Range likely to contain the true mean; excluding 0 means significant.

Mismatch floor
Residual drift (about 3 MHz) left by chip-to-chip mismatch that a global correction cannot remove.

H1
Hypothesis that residual error scales as $C\times$ raw error; needs Spectre, pending.